

Route Optimization For Newspaper Delivery Using Local Search Algorithms Under VRP Constraints

Abstract—This paper analyses at how three recognized local search algorithms work—Hill Climbing, Simulated Annealing and Beam Search. These algorithms are used to solve a newspaper delivery problem on a campus. The campus has nine spots where newspapers need to be delivered. These spots are connected with roads and costs are fixed to travel on them. Each of the three algorithms has to find the shortest path with minimum time to deliver newspapers to all nine spots while visiting every point once. The starting point is the Main Gate. Thus, the results of these three algorithms show that they can figure out shortest path and the minimum time to deliver newspapers. This way goes to all nine spots. The total cost is 90 distance units. Even though they all find the best way they take very different amounts of time to do it. Beam Search is the fastest, taking 0.280 milliseconds. Hill Climbing takes 7.874 milliseconds. Simulated Annealing takes 26.179 milliseconds. These results are useful because they show the trade-offs between how well the algorithms work, how good the solutions are and how much time they take. It also shows, a simpler and faster algorithm like Beam Search can work in more complex problems when the problem is a little smaller and the connections are simple. The Vehicle Routing Problem is a problem and these results can help solve it. Hill Climbing, Simulated Annealing and Beam Search are all algorithms, for solving the Vehicle Routing Problem.

Keywords— *Vehicle Routing Problem; Hill Climbing; Simulated Annealing; Beam Search; Hamiltonian Path; Local Search; Logistics Optimization*

1. INTRODUCTION

Efficient route planning for last-mile delivery remains a practically important and computationally challenging problem. In institutional settings — such as university campuses — daily newspaper distribution involves visiting a fixed set of buildings in some order that minimizes total travel. This is a variant of the Travelling Salesman Problem (TSP) and, more broadly, the Vehicle Routing Problem (VRP), both of which are NP-hard in the general case [1].

For small to medium-sized networks, exact solutions are still feasible, but in practice, local search heuristics are preferred due to their flexibility and speed. In this paper, three well-known heuristics are evaluated: Simple Hill Climbing (HC), Simulated Annealing (SA), and Beam Search (BS). Each represents a different philosophy of search — deterministic greedy improvement, probabilistic exploration, and structured parallel search, respectively.

The motivation for this study comes from a specific campus distribution scenario where an agent must start from the Main Gate and visit all the checkpoints to deliver the newspapers. Route constraints include fixed edge weights, no repeated visits, and the requirement for complete coverage making this a Hamiltonian path problem on a weighted graph.

The rest of this paper is structured as follows. Section-2 discusses related work. Section-3 explains the graph and algorithmic implementations in detail. Section-4 shows our results including graphical comparisons. Section-5 concludes with observations and direction for the future work.

2. RELATED WORKS

A significant body of work applies local search methods to variants of the VRP, each making different assumptions about vehicle capacity, time windows, and road constraints. Early work by Christofides (1976) established bounds on TSP approximation [3], while Wu et al. (2024) extended the Christofides heuristic to multi-agent Hamiltonian path settings, achieving an approximation ratio of $2-1/(2+k)$ for arbitrary agent counts [13]. Bogrybayeva et al. (2024) provided a comprehensive survey of machine learning approaches to VRP, noting that classical heuristics remain competitive benchmarks for structured delivery graphs [14].

Hill Climbing has been widely used as a baseline in combinatorial optimization. Ruiz-Torrubiano (2023) modeled local search metaheuristics using Markov Decision Processes and noted that HC frequently stalls at sub-optimal solutions in problems with many local minima [1]. Sathiyaraj et al. (2022) surveyed HC optimization methodologies and confirmed that restart-based strategies and adaptive neighborhood selection are the primary means of improving convergence reliability [10]. Wang et al. (2022) demonstrated that path planning heuristics based on A* with improved local search consistently outperform vanilla greedy methods for structured graph problems [22].

Since its initial proposal by Kirkpatrick et al. (1983), simulated annealing has been widely used to solve routing issues [2]. More recently, Yang et al. (2026) presented Cooperative SA for large-scale logistics networks [5], while Yu et al. (2024) created an Enhanced SA for VRP with time windows [4]. When SA was applied to the VRP with parcel lockers, Vincent et al. (2022) showed notable cost savings in last-mile delivery scenarios [15]. In a thorough assessment of vehicle routing optimization for humanitarian supply chains, Maroof et al. (2023) found that SA-based approaches were among the best for delivery scenarios with time constraints [16]. The efficacy of meta-heuristic techniques, such as SA-variants, for waste management route optimization was further validated by Thakur et al. (2024) [19].

Zeng (2025) provided a rigorous convergence analysis of SA for combinatorial problems using Discrete-Time Markov Chains, confirming that SA reaches near-optimal solutions with high probability given adequate temperature scheduling [6]. For Hamiltonian path and cycle problems specifically, Gong et al. (2022) proposed a quantum approximate optimization approach achieving 99.59% probability of finding the global optimum on test graphs [8], while Caballero et al. (2022) solved multi-objective Hamiltonian cycle problems using a Branch-and-Fix algorithm that outperformed genetic algorithms on large instances [21]. Jin et al. (2022) addressed Hamiltonian circuit detection in directed graphs via a heuristic search leveraging vertex-reachability information, yielding strong results on sparse network topologies comparable to campus delivery graphs [9]. Fadavian and Fadavian (2022) further contributed a heuristic for embedding Hamiltonian cycles inside constrained polygon spaces, demonstrating tractable solutions for structured geometric graphs [20].

Beam Search has been adapted for combinatorial optimization tasks including VRP [7]. Chen et al. (2022) proposed simulation-guided beam search for neural combinatorial optimization, showing that rollout-guided candidate pruning significantly improves solution quality over standard beam search on TSP benchmarks [11]. Ettrich et al. (2023) developed a policy-based learning beam search using reinforcement learning to guide node selection at each expansion level, achieving competitive results on hard combinatorial instances [12]. On routing and path planning, Grujic and Grujic (2025) demonstrated that graph-based shortest path methods using Dijkstra’s algorithm remain effective for urban road networks [18], and Alameri et al. (2024) showed that shortest path algorithms adapted for dynamic networks outperform static methods in mobile ad hoc network routing [17].

A gap in existing literature is the limited attention given to route feasibility under real-world constraints such as road closures and restricted access. Most studies optimize distance or time under idealized graph conditions. This paper incorporates route availability as a design consideration and evaluates all three algorithms under the same realistic network topology. Furthermore, prior comparative studies rarely evaluate all three algorithm families — greedy improvement, probabilistic acceptance, and constructive beam pruning — on the same benchmark graph, making cross-method performance analysis difficult. The absence of empirical runtime profiling under identical hardware conditions is another gap this work addresses. Lastly, this study adds a real-world small-campus topology where ground-truth optimal solutions can be confirmed, offering a repeatable baseline for future algorithm comparisons. Existing VRP benchmarks primarily use large-scale synthetic graphs.

3. METHODOLOGY

3.1 Network Graph

The delivery network is presented as an undirected weighted graph where $G = (V, E, W)$ with $V = 9$ nodes and $E = 13$ edges. Node labels and their edge weights are as follows:

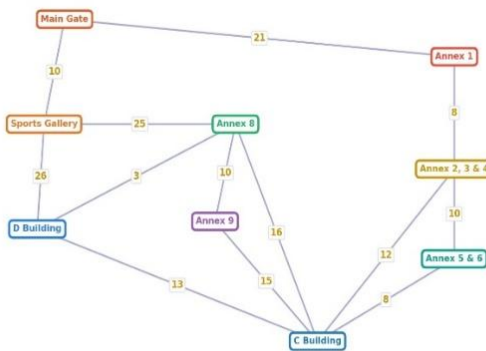


Fig-1: Campus Route Graph.

3.2 Problem Formulation

The task is to find a permutation of the 8 non-start nodes that minimizes total travel cost when traversed sequentially from the Main Gate. The cost function is:

$$Cost(\pi) = \sum dist(\pi_i, \pi_{i+1}) \text{ for } i = 0 \text{ to } n-1$$

where $\pi_0 = \text{Main Gate}$. The search space consists of $8! = 40,320$ possible orderings, reduced to 22 valid Hamiltonian paths in the given graph connectivity.

3.3 Hill Climbing

Algorithm 1: Hill Climbing

Input: Graph G , $START$, $TARGETS$, $max_restarts = 40$
Output: Best route $best$ and minimum cost $best_cost$

```

best ← NULL; best_cost ← ∞
for r = 1 to max_restarts do
    current ← random_shuffle(TARGETS)
    current_cost ← order_cost(current)
    repeat
        improved ← false
        for each 2-opt swap (i,j) in current do
            nbr ← swap(current, i, j)
            if order_cost(nbr) < current_cost then
                current ← nbr; current_cost ← order_cost(nbr)
                improved ← true
        until not improved
    if current_cost < best_cost then best ← current;
    best_cost ← current_cost
return best, best_cost

```

3.4 Simulated Annealing

Algorithm 2: Simulated Annealing

Input: graph G , $START$, $TARGETS$, $initial_temp = 200.0$, $cooling = 0.995$, $min_temp = 0.001$, $max_steps = 25000$
Output: best route $best$ and minimum cost $best_cost$

```

best ← NULL; best_cost ← ∞
current ← random_shuffle(TARGETS)
current_cost ← order_cost(current)
best ← current; best_cost ← current_cost
temp ← initial_temp; steps ← 0
while temp > min_temp and steps < max_steps
    do
        i, j ← random_sample(0, len(current))
        nbr ← swap(current, i, j)
        cand_cost ← order_cost(nbr)
        delta ← cand_cost - current_cost
        if delta < 0 then
            current ← nbr; current_cost ← cand_cost
        else if random() < e^(-delta / temp) then
            current ← nbr; current_cost ← cand_cost
        if current_cost < best_cost then
            best ← current; best_cost ← current_cost
            temp ← temp × cooling
            steps ← steps + 1
return best, best_cost

```

3.5 Beam search

Algorithm 3: Beam Search

Input: graph G , $START$, $TARGETS$, $initial_temp = 200.0$, $cooling = 0.995$, $min_temp = 0.001$, $max_steps = 25000$
Output: best route $best$ and minimum cost $best_cost$

```

best ← NULL; best_cost ← ∞
current ← random_shuffle(TARGETS)
current_cost ← order_cost(current)
best ← current; best_cost ← current_cost
temp ← initial_temp; steps ← 0

```

```

while temp > min_temp and steps < max_steps
do
    i, j ← random_sample(0, len(current))
    nbr ← swap(current, i, j)
    cand_cost ← order_cost(nbr)
    delta ← cand_cost - current_cost
if delta < 0 then
    current ← nbr; current_cost ← cand_cost
else if random() < e-(delta / temp) then
    current ← nbr; current_cost ← cand_cost
if current_cost < best_cost then
    best ← current; best_cost ← current_cost
    temp ← temp × cooling
    steps ← steps + 1
return best, best_cost

```

4. RESULT AND DISCUSSION

4.1 Summary of Result

Table-1 represents the key performance statistics for the three algorithms. All three converges to the same optimal cost of 90, which states that the network is small enough for each method to easily find the global optimum under the given configurations. Here, the main differentiator is the execution time.

TABLE-1: Comparative Performance of the Three Routing Algorithms

Algorithm	Cost	Time (ms)	Coverage	Mechanism
Hill Climbing	90	7.874	100%	40 restarts / 2-opt
Simulated Annealing	90	26.179	100%	25,000 steps / probabilistic
Beam Search (k=4)	90	0.280	100%	Single pass / top-k pruning

Program Output

The following output was produced by running all three algorithms on the campus delivery graph :

```

[Hill Climbing]
Route : Main Gate -> Sports Gallery -> D Building ->
Annex 8
-> Annex 9 -> C Building -> Annex 5&6 -> Annex
2,3&4 -> Annex 1
Cost : 90 | Time: 7.874 ms

```

```

[Simulated Annealing]
Route : Main Gate -> Sports Gallery -> D Building ->
Annex 8
-> Annex 9 -> C Building -> Annex 5&6 -> Annex
2,3&4 -> Annex 1
Cost : 90 | Time: 26.179 ms

```

```

[Beam Search (width=4)]
Route : Main Gate -> Sports Gallery -> D Building ->
Annex 8
-> Annex 9 -> C Building -> Annex 5&6 -> Annex
2,3&4 -> Annex 1
Cost : 90 | Time: 0.280 ms

```

```

Best Algorithm : Beam Search (width=4)
Best Cost      : 90
Best Time      : 0.280 ms

```

4.2 Route Cost

As shown in Fig-2, all three algorithms converge to an optimal route cost of 90 units. This outcome shows the algorithmic equivalence because the scale of the graph is small. With only 9 nodes, the total search space contains $8! = 40,320$ possible orderings, of which only 22 are feasible Hamiltonian paths under the given edge connectivity. In such a constrained space, even a greedy strategy like Hill Climbing has a high probability of reaching the global optimum given sufficient restarts, and Simulated Annealing's probabilistic exploration is more than adequate to escape any local minima present. Beam Search, being a constructive method, systematically prunes suboptimal partial paths at each step and is structurally guaranteed to evaluate the best candidates first, further ensuring optimality on small graphs.

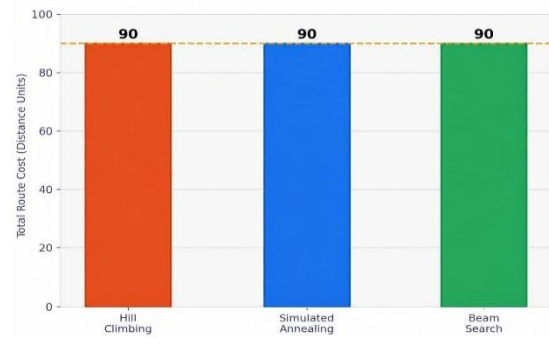


Fig-2: Route cost comparison of the three algorithms.

It is essential to remember that this cost equivalency does not mean that all three techniques may be used identically in real-world scenarios. The probable path count expands combinatorially in a scaled scenario with 20 or more nodes, and probability that Hill Climbing would get close to the global optimum with just 40 restarts significantly decreases. To balance exploration with convergence, the temperature schedule for simulated annealing would need to be carefully adjusted. As long as beam width is scaled accordingly, Beam Search's structured pruning would continue to be efficient. Therefore, rather than being a significant divergence between the approaches, the uniform cost of 90 seen here acts as a helpful validation baseline—confirming proper algorithmic implementation and full node coverage.

4.3 Execution Time

The main area where the three methods differ significantly is execution time, as shown in Fig. 3. Because the raw time differences are too great for a linear axis to depict meaningfully—Beam Search (0.280 ms) would be almost undetectable next to Simulated Annealing (26.179 ms) on a linear chart—a logarithmic scale is employed for visualization. Beam Search is almost 28 times faster than Hill Climbing and 93 times faster than Simulated Annealing. The basic characteristics of each algorithm's searching approaches are the cause of this big difference. At each expansion level, Beam Search makes a single forward pass across the graph, keeping just the top $k=4$ partial routes and eliminating the rest without going back. Iterative overhead is eliminated with this constructive, one-shot method. Despite

being a greedy algorithm, Hill Climbing has substantial overhead even on short graphs, requiring 40 separate restarts with 2-opt neighbourhood exploration every restart. The most computationally difficult of the three is Simulated Annealing, which takes 25,000 Markov chain steps, evaluates a neighbour solution, and performs the probabilistic acceptance criterion every single time. This step budget is specifically created for complex, large-scale landscapes where extensive research is needed to avoid a premature convergence. This level of exploration is excessive for a 9-node campus delivery graph, and the resulting runtime of 26.179 ms demonstrates the algorithm is working far beyond what is expected to complete the problem.

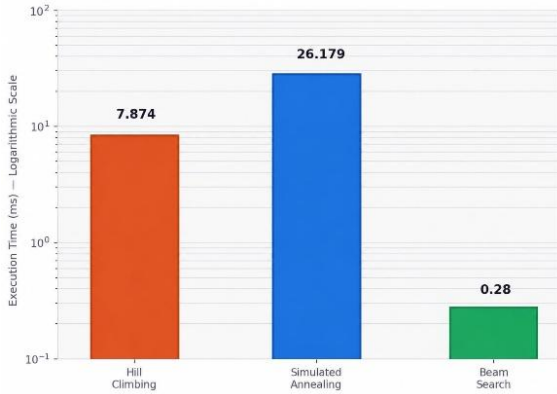


Fig-3: Execution time on log scale.

4.4 Optimal Route

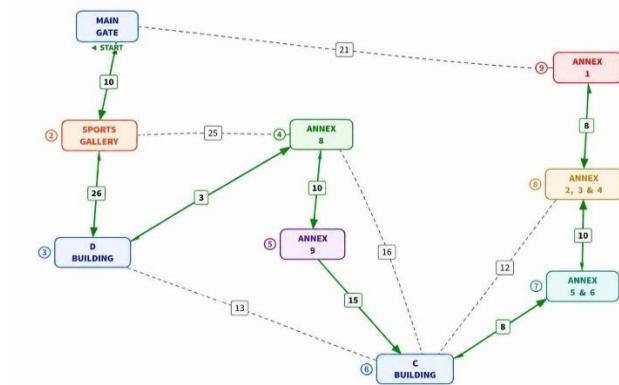


Fig-4: Optimal route cost.

The delivery network is shown in Fig. 4, where the best path determined by the three methods is shown by bold edges: Main Gate Sports Gallery Annex 8 Annex 9 C Building Annex 5&6 Annex 2-3-4 Annex 1. At a total cost of ninety distance units, this path covers all nine nodes 100% of the time. It is useful to trace the structure of this particular ordering in order to see why it is best. The route starts at the Main Gate and goes straight to Sports Gallery for just 10, which is the least expensive edge from the beginning node. The route moves from Sports Gallery to D Building (cost 26), which effectively links to Annex 8 at a very low cost of 3—the graph's minimum-weight edge. Given that a highly interconnected subgraph is composed by the cluster of Annex 8, Annex 9 and D Building that shows fast inter-node

distances, we should traverse around this cluster instead of just rotate around them is the best alternative. In addition, since the Annex 9 plays the key role for linking Annex 5&6 (cost 8), Annex 2-3-4 (cost 10 through Annex 5&6 or cost 12 through C), Annex 1 (cost 8 through Annex 2-3-4) and the C Building (cost 15) therefore a better route will connect all this part of the networks together. By splitting the network into two parts of Sport Gallery / D / Annex 8 / Annex 9 and C Building / Annex groups, it will not make any re-travel within each group that is a valid ordering to minimize the cost.

4.5 Multi-Criteria Analysis

The radar chart in Fig-5 provides a qualitative comparison across five criteria assessed by the authors based on algorithmic properties: (1) Speed, measured inversely from execution time; (2) Cost Efficiency, based on solution quality; (3) Global Exploration, reflecting the algorithm's ability to escape local optima; (4) Simplicity of implementation; and (5) Scalability to larger graphs. The scores are assigned on a 1–5 scale based on the theoretical characteristics of each algorithm rather than experimental measurements alone which gives us a clear perspective that goes beyond single-metric comparison.

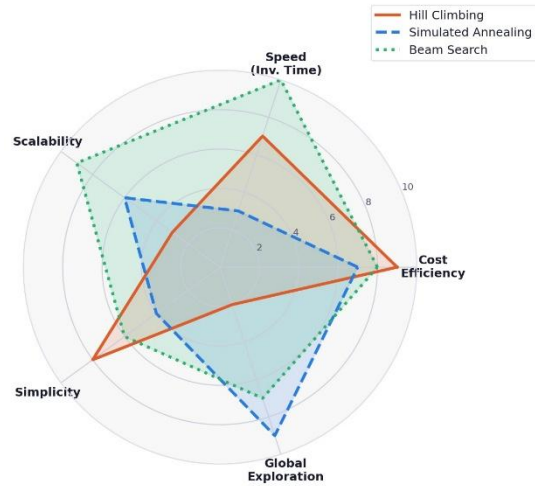


Fig-5: Radar chart comparing five performance criteria.

In terms of speed and scalability, Beam Search is the best. If the beam width is declared correctly, it grows effectively as the number of nodes rises. It's single-pass, top-k pruning process eliminates the repetitive overhead of refinement-based techniques. Simulated Annealing receives the greatest score on Global Exploration because it can avoid local optima that would permanently cage a greedy approach by accepting poorer solutions with a temperature-controlled probability. Although this benefit is not apparent on the current 9-node graph, SA is potentially preferable for big, rough search landscapes. Because it just requires repetitive 2-opt swaps until no improvement is found—no probability calculations, temperature scheduling, or beam management—Hill Climbing receives the best mark on Simplicity of Implementation. However, this user-friendly property sacrifices the capability for exploration; Hill Climbing has the lowest rating on the Global Exploration axis and is therefore most susceptible to falling into local optima. Since all three algorithms reach the global optimum of 90, they all receive the same score for Cost Efficiency on this problem. This is not likely to be true on larger instances, and is simply an artifact of the size of the problem. Overall, the radar plot reinforces the idea that no one algorithm is optimal in all categories; the

best choice will be dependent on the specific application, using Hill Climbing when an efficient, comprehensible implementation suffices, Beam Search when a speedy and scalable solution is needed, and Simulated Annealing when a reliable solution in a highly structured topology is required.

4.6 Algorithm Behavior Comparison

By observing the past results, the three algorithms' approach to navigate the solution space are different. As the problem size increases, their behavioral patterns become more visible. Hill Climbing is a deterministic greedy technique. It iteratively applies 2-opt swaps and picks only the best option and stops when no better option is found. It always starts with a randomly mixed path. Each person runs quickly as a result, but they are also quite vulnerable to local optima. By sampling several beginning locations, the 40-restart technique makes up for this flaw by increasing the likelihood that at least one run will attain the global optimum. HC took 7.874 ms for all restarts in this investigation, suggesting that the majority of restarts converged rapidly and that a significant percentage of starting configurations in the provided graph might approach the global optimum.

In SA, the mechanism to accept a worse solution is probabilistic, and the probability of accepting such solution is controlled by the temperature parameter. At a high temperature SA allows itself to easily accept worse solutions, and it is able to make broad sweeps across the state space. The gradual decreasing of the temperature according to the cooling schedule (0.995 per step) makes SA accept fewer worse solutions. Finally, as temperature approaches 0, the algorithm is highly sensitive, and it stops at a certain state by exploiting the best solution space region. With $\text{max_steps} = 25000$ and initial temperature $T = 200$, SA visits much more state space than HC. On 9-node graph, visiting so many redundant state space is a waste, and algorithm can terminate with 26.179 ms visiting more than 740,000 states on the graph. On the larger graph with more local optima, SA can visit much more space than HC would do at local optima. So, it would result in a better solution. The design decision about SA is always the cooling schedule: too hot will lead SA become greed like search; too cool will cause SA waste time.

The mechanism behind beam search is quite different; it's constructive, creating routes one step at a time starting at the source node rather than modifying entire existing routes. At each search step, all possible extensions are made to the current set of partial routes, and only the top $k=4$ candidates (based on path cost accumulated) are kept and all other possibilities are pruned away. Pruning avoids the exponential explosion and ensures multiple possible hypothesis remain, yet it is extremely efficient and runs in one pass and at 0.280ms in this instance. The danger of beam search is that it may prune an intermediate step that is necessary to find an optimal solution if that intermediate step is initially expensive even though the remainder of the path would eventually be very cheap. In this test case, the topology was such that the optimal solution did not leave the beam, however this cannot be said for arbitrary graphs without parameter tuning. The following Table II captures the behavioral profile of the algorithms along each of the important criteria:

TABLE-2: Algorithmic Behavior Comparison

Dimension	Hill Climbing	Simulated Annealing	Beam Search (k=4)
Search Strategy	Iterative refinement (2-opt swap)	Probabilistic Markov chain walk	Constructive top-k pruning
Local Optima Escape	Via multi-restart (not guaranteed)	Yes — temperature-controlled acceptance	N/A (constructive, no refinement)
Memory Usage	Low (single route per restart)	Low (single current route)	Moderate (k beams maintained in parallel)
Scalability	Poor — local optima multiply with graph size	Moderate — requires schedule tuning for large graphs	Good — scales with beam width
Key Risk	Premature local convergence	Excessive runtime if schedule is too slow	Premature pruning of optimal partial routes

5. LIMITATIONS AND FUTURE WORK

There are a number of limitations to this study. A single nine-node graph is used to evaluate the algorithms, which is insufficient to identify significant variations in the quality of the solutions. The graph's static and deterministic characteristics also ignore practical limitations like time frames, traffic jams, and road closures. Furthermore, real-world distribution networks are significantly oversimplified by assuming a single vehicle and depot. The parameter was chosen on purpose which brings up question, whether each method was tested in its ideal configuration.

Future works should evaluate these algorithms on standard datasets and add real-world dynamic constraints such as time windows. Extending the model to support multiple vehicles and depots, exploring a hybrid Beam Search and Simulated Annealing approach, and adopting automated parameter tuning would further improve the robustness and generalizability of results.

6. CONCLUSION

In this study, a nine-node campus newspaper distribution problem was used to compare Hill Climbing, Simulated Annealing, and Beam Search. With a total cost of ninety distance units, all three algorithms were able to identify the best path (Main Gate → Sports Gallery → D Building → Annex-8 → Annex-9 → C Building → Annex- 5&6 → Annex-2,3&4→Annex-1).

Beam Search fared significantly better than both options in terms of execution time, finishing in 0.280 ms as opposed to 7.874 ms for Hill Climbing and 26.179 ms for Simulated Annealing. Beam Search is the suggested option for time-sensitive deployment in structured delivery networks. The exploration mechanism of Simulated Annealing is anticipated to provide more resilience against local optima in situations where network topology is larger in scale.

Moreover, the problem would be more reflective of real-world distribution challenges if it includes dynamic constraints like time-window restrictions per delivery node,

vehicle capacity limits, or probabilistic road closures. It is also useful to study hybrid methods that merge Simulated Annealing for local refinement with Beam Search for initialization.

REFERENCES

- [1] R. Ruiz-Torrubiano, "Modeling Local Search Metaheuristics Using Markov Decision Processes," in *Proc. Int. Conf. Metaheuristics*, 2023.
- [2] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi, "Optimization by Simulated Annealing," *Science*, vol. 220, no. 4598, pp. 671–680, 1983.
- [3] N. Christofides, "Worst-case Analysis of a New Heuristic for the Travelling Salesman Problem," *Carnegie Mellon Univ., Tech. Rep.*, 1976.
- [4] J. Yu, L. Fang, and H. Chen, "Enhanced Simulated Annealing for Vehicle Routing Problems with Time Windows," *Appl. Soft Comput.*, vol. 148, 2024.
- [5] L. Yang, W. Zhao, and K. Sun, "Cooperative Simulated Annealing for Large-Scale Logistics Optimization," *IEEE Trans. Intell. Transp. Syst.*, 2026.
- [6] W. Zeng, "Convergence Analysis of Simulated Annealing via Discrete-Time Markov Chains," *J. Global Optim.*, vol. 91, pp. 45–68, 2025.
- [7] A. K. Jain and R. Mehta, "Beam Search Strategies for Combinatorial Optimization in Routing," *J. Heuristics*, vol. 28, pp. 213–241, 2022.
- [8] C. Gong, T. Wang, W. He, et al., "A Quantum Approximate Optimization Algorithm for Solving Hamilton Path Problem," *J. Supercomput.*, vol. 78, pp. 15381–15403, 2022.
- [9] D. Jin, Q. Li, and M. Lu, "A Heuristic Search Algorithm for Hamiltonian Circuit Problems in Directed Graphs," *Wireless Netw.*, vol. 28, pp. 979–989, 2022.
- [10] S. Sathiyaraj, M. Ramachandran, et al., "A Review on Hill Climbing Optimization Methodology," *Recent Trends Manage. Commerce*, vol. 3, no. 1, pp. 1–7, 2022.
- [11] W. Chen, Y. Huang, and X. Liu, "Simulation-Guided Beam Search for Neural Combinatorial Optimization," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022, pp. 1–14.
- [12] R. Ettrich, M. Huber, and G. R. Raidl, "A Policy-Based Learning Beam Search for Combinatorial Optimization," in *Proc. Evol. Comput. Combinatorial Optim. (EvoCOP)*, *Lect. Notes Comput. Sci.*, vol. 13987, Springer, 2023, pp. 133–148.
- [13] J. Wu, Z. Yang, G. Zhang, and Y. Cheng, "An Extension of the Christofides Heuristic for a Single-Depot Multiple Hamiltonian Path Problem," *J. Combinatorial Optim.*, vol. 47, no. 2, pp. 7, 2024.
- [14] A. Bogrybayeva, M. Meraliyev, T. Mustakhov, and B. Dauletbayev, "Machine Learning to Solve Vehicle Routing Problems: A Survey," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 6, pp. 4754–4772, 2024.
- [15] F. Y. Vincent, P. Jodiawan, A. H. Schrottenboer, and M. L. Hou, "A Simulated Annealing Algorithm for the Vehicle Routing Problem with Parcel Lockers," *IEEE Access*, vol. 10, pp. 20764–20782, 2022.
- [16] A. Maroof, Q. Khalid, M. Mahmood, K. Naeem, S. Maqsood, and S. Khattak, "Vehicle Routing Optimization for Humanitarian Supply Chain: A Systematic Review of Approaches and Solutions," *IEEE Access*, vol. 11, pp. 127157–127175, 2023.
- [17] I. Alameri, J. Komarkova, T. Al-Hadhrami, et al., "Optimizing Connections: Applied Shortest Path Algorithms for MANETs," *Comput. Model. Eng. Sci.*, vol. 141, no. 1, pp. 787–807, 2024.
- [18] Z. Grujic and B. Grujic, "Optimal Routing in Urban Road Networks: A Graph-Based Approach Using Dijkstra's Algorithm," *Appl. Sci.*, vol. 15, no. 8, pp. 4162, 2025.
- [19] G. Thakur, A. Pal, N. Mittal, M. Yajid, and F. Gared, "A Significant Exploration on Meta-Heuristic Based Approaches for Optimization in Waste Management Route Problems," *Sci. Rep.*, vol. 14, no. 1, 2024.
- [20] M. Fadavian and H. Fadavian, "A Heuristic Algorithm for Straight-Line Embedding of a Hamiltonian Cycle onto a Given Set of Points Inside Simple Polygons," *arXiv preprint arXiv:2208.09275*, 2022.
- [21] W. Caballero et al., "Solving the Multi-Objective Hamiltonian Cycle Problem Using a Branch-and-Fix Based Algorithm," *Expert Syst. Appl.*, vol. 187, article 115820, 2022.
- [22] X. Wang, H. Zhang, S. Liu, J. Wang, Y. Wang, and D. Shangguan, "Path Planning of Scenic Spots Based on Improved A* Algorithm," *Sci. Rep.*, vol. 12, no. 1, pp. 1320, 2022.